Open-economy policy and regimes

How does currency adjustment interact with demand and policy constraints?

Today's route

Trade prices and quantities determine the spending effect of FX.

DD–AA connects output with asset-market equilibrium.

Pegs, floating rates and the trilemma change the policy experiment.

LECTURE 5

Trade prices and quantities

Does a depreciation automatically improve the trade balance?

Exporters and importers face different cash flows



Invented trading economy. Imported inputs can be essential to domestic production.

The real exchange rate and net exports

$$q = \frac{EQ^{\text{foreign}}}{Q^{\text{home}}}, \qquad NX = NX(q, Y, Y^{\text{foreign}}).$$

A higher q makes home goods relatively cheaper.

Higher home income can raise imports.

A depreciation raises the local import bill first

Balanced trade initially	Before	Immediate effect
Real exchange rate q	1.00	1.10
Exports, fixed quantity and price	100	100
Import quantity, fixed	100	100
Home value of imports	100	110
Net exports	0	-10

Producer-currency prices are fixed. Quantities have not adjusted.

Quantities can later reverse the result

After the same depreciation	Value
Export volume rises 8%	108
Import volume falls 6%	94
Import value: 1.10×94	103.40
Net exports: 108 - 103.40	+4.60

Invented quantity responses, after enough time to adjust.

Marshall–Lerner gives a local condition

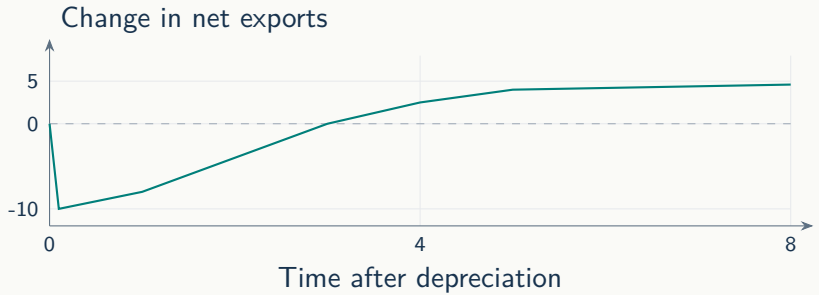
$$\epsilon_X + \epsilon_M > 1.$$

ϵ_X: export-volume elasticity to real depreciation.

ϵ_M: absolute import-volume elasticity.

Balanced starting trade and fixed producer prices in the simplest derivation.

The J-curve is a timing mechanism



Original hypothetical path: the import price effect precedes quantity adjustment.

Invoice currency changes pass-through

Euro exporter to a dollar buyer	Initial	After euro depreciation
USD per EUR	1.10	1.00
EUR invoice: EUR 100 in USD	110	100
USD invoice: USD 110 in USD	110	110

Same exchange-rate move. Different invoice contracts. Hypothetical prices.

The dollar can matter in third-country trade

A UK importer buys from Korea at an invoice price of USD 100.

GBP per USD rises from 0.80 to 0.88.

The sterling bill rises from GBP 80 to GBP 88.

LECTURE 5

The DD schedule

Which output and exchange-rate pairs balance the goods market?

A short-run open-economy model

$$Y = C(Y - T) + I + G + NX(q, Y, Y^{\text{foreign}}).$$

Goods prices, foreign conditions and expected future FX are fixed.

For this experiment, investment I is fixed too.

A numerical demand equation

$$D = 40 + 0.5Y + 10E.$$

D: planned spending on home output.

A higher E raises demand at each output level.

Only half of extra output returns as extra planned spending.

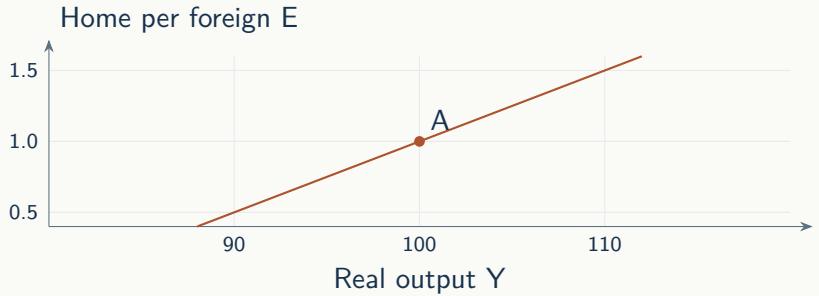
Goods-market equilibrium gives DD

$$Y = D \quad \Rightarrow \quad Y = 80 + 20E.$$

A weaker currency raises demand and equilibrium output.

DD slopes upward with Y horizontal and E vertical.

DD as a graph



Original numerical goods-market equilibrium: $Y = 80 + 20E$.

LECTURE 5

The AA schedule

What second relationship pins down output and the currency?

Money demand and FX pricing together

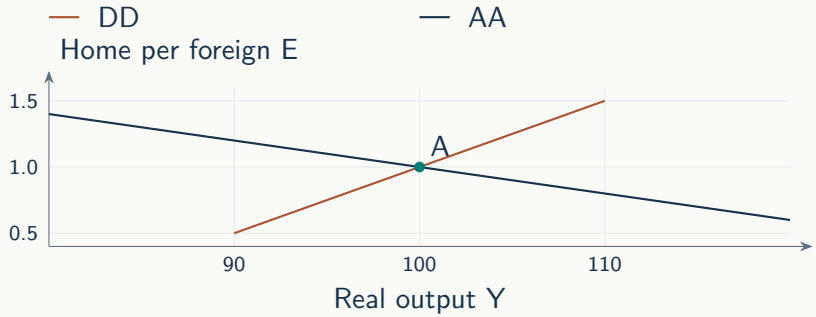
$$M/Q = L(Y, R^{\text{home}}).$$

$$R^{\text{home}} \approx R^{\text{foreign}} + \mathbb{E}_t \Delta \ln E.$$

Higher Y raises money demand and the home rate.

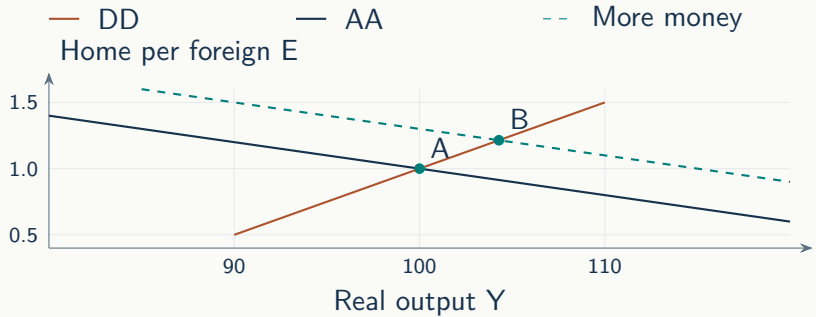
At fixed expected future FX, that supports home appreciation.

AA slopes downward



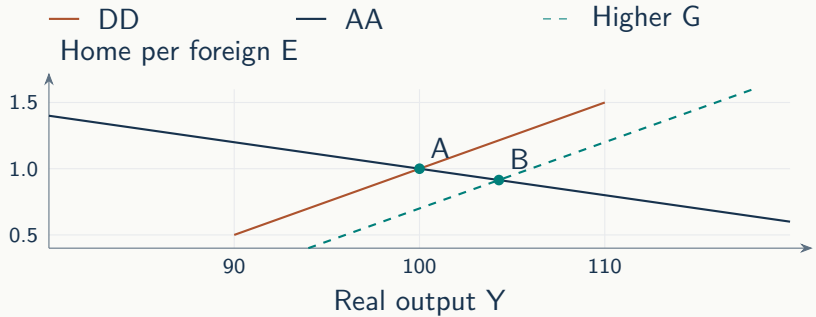
The intersection A balances goods and asset markets. Original linear illustration.

Temporary monetary expansion under a float



AA shifts up. In this model output rises and the home currency depreciates.

Temporary fiscal expansion under a float



DD shifts right. Output rises, while currency appreciation offsets some demand.

A fiscal expansion need not improve net exports

Government demand rises directly.

Higher income can increase imports.

Appreciation can reduce expenditure switching toward home goods.

At the lower bound, expectations still matter

Cash limits how far nominal rates can be pushed below zero.

Extra reserves may have little effect on the current short rate.

Credible changes in future policy can still affect real rates and FX.

LECTURE 5

Pegs and interventions

What changes when the central bank promises a fixed currency price?

The promise behind a peg

The central bank stands ready to exchange currencies at the peg.

It must supply foreign reserves when investors want to exit.

The promise constrains monetary policy when capital moves freely.

Defending a peg changes the balance sheet

Central bank transaction	Assets	Liabilities
Sell 10 of foreign reserves	Foreign assets -10	
Receive home reserve balances		Bank reserves -10

An unsterilised sale drains domestic liquidity.

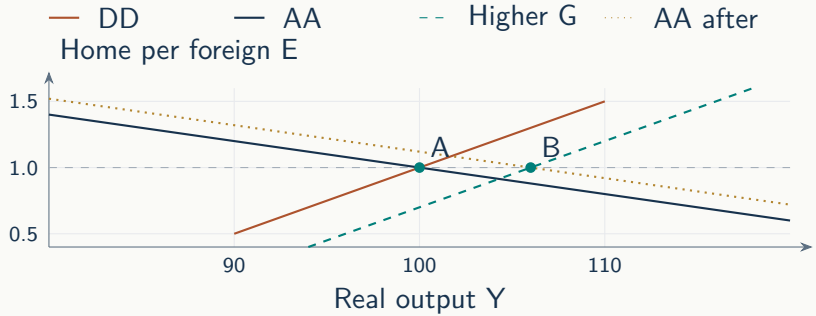
Sterilisation offsets the liquidity effect

Second transaction	Assets	Liabilities
Buy 10 of domestic bonds	Domestic assets +10	Bank reserves +10

Total monetary liabilities return to their initial size.

The asset portfolio now contains fewer foreign and more domestic assets.

A fixed peg removes one degree of freedom



At E = 1, higher G raises output from 100 to 106 if monetary policy accommodates.

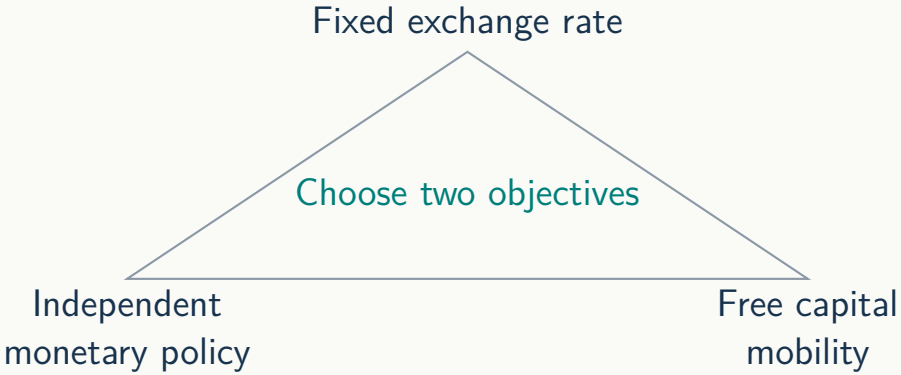
Independent monetary expansion conflicts with the peg

A rate cut creates pressure for home-currency depreciation.

Defending the peg withdraws liquidity or reverses the rate cut.

With an unchanged credible peg, the independent impulse cannot persist.

The monetary trilemma



LECTURE 5

Gold and the international system

What does a monetary anchor buy, and what adjustment does it impose?

Gold convertibility



Illustration of redemption, not a historical photograph or certificate.

Gold fixes a currency relationship

$$1 \text{ home unit} = a \text{ grams}, \qquad 1 \text{ foreign unit} = b \text{ grams}.$$

$$E = \frac{b}{a} \quad \text{home units per foreign unit}.$$

Transport, insurance and conversion costs create a band around parity.

Deflation makes a fixed nominal rate tighter

$$R^{\text{nom}} = 0\%, \quad \mathbb{E}\pi = -5\% \quad \Rightarrow \quad R^{\text{real}} \approx 5\%.$$

Falling expected prices can raise the real cost of borrowing.

A monetary anchor can constrain the response.

Three monetary arrangements

Arrangement	Anchor	Adjustment problem
Gold standard	Gold convertibility	Domestic demand may bear the strain
Bretton Woods	Dollar pegs and dollar-gold link	Reserve demand versus convertibility
Floating fiat currencies	Domestic policy frameworks	FX absorbs shocks, can also amplify them

Triffin's dilemma

Global demand for dollar reserves requires dollar liabilities.

More claims relative to gold can weaken confidence in convertibility.

Providing liquidity and preserving a fixed gold promise can conflict.

Counterfactual exercise: who can ease?

Economy	Regime	Can cut independently?
A	Float, mobile capital	?
B	Credible peg, mobile capital	?
C	Peg, effective capital controls	?

Assume the foreign policy rate stays fixed and no devaluation is expected.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.